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NPS/001/016 – Technical Specification for Compression and Mechanical Fittings for Overhead Lines

1. Purpose

The purpose of this document is to detail the technical requirements for compression and mechanical fittings used on the CE Electric distribution networks.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/004/015	1.0	Technical Specification for Compression Fittings for Low Voltage Overhead Lines
DSS/004/016	1.0	Technical Specification for Compression Fittings for ACSR Overhead Lines
DSS/005/005	1.0	Technical Specification for Line Taps for Copper Conductors
MS – O/09		Specification for Compression Fittings for Overhead Line
MS – O/33		Specification for Split Bolt Connectors for Overhead Lines
MS – O/25		Specification for Live Line Tap Fittings for Overhead Lines
NPS/001/016	1.0	Specification for Compression and Mechanical Fittings for Overhead Lines

2. Scope

This specification applies to all compression and mechanical fittings for use on the CE Electric UK overhead line distribution network and details their range and size where applicable. The majority of items are detailed in national, international and industry standards that are listed within the technical description for each product type.

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3. Technical Requirements

3.1 General

ENA TS 43-92 Issue 4, 2004 details the standard to which overhead line compression and mechanical fittings shall generally be supplied.

Materials used to manufacture fittings shall be able to withstand the mechanical load conditions relevant to their installation and environmental conditions. They must also be compatible with and have no deleterious effect on the conductor.

All ferrous materials shall be protected against corrosion by hot dip galvanising to the requirements of BSEN 1461. The fitting material should not be adversely affected by any coating applied for corrosive protection.

Tolerances shall be in accordance with ENA TS 43-92 Issue 4 Section 4.2

3.2 Compression Conductor Fittings

Compression fittings shall be range taking where possible and designed to cover a standard range of overhead line conductors. The items include tension joints and terminations, non-tension joints, tee-joints and bi-metal configurations are detailed in Appendix B.

Fittings shall be marked in accordance with ENA TS 43-92 Issue 4 Sections 4.3 and 5.2

Compression fittings shall be suitable for application using the universal die-less tooling system currently used within CE Electric UK, except specified ACSR and AAAC fittings detailed in Appendix B.

All components shall be fitted with approved grit grease to prevent metal oxidation, provide electrical conductivity and provide tensile holding properties. The connector orifice shall be fitted with a plastic cap or plug.

The majority of tubular fittings shall have a conductor stop at the centre of the joint. The only exception is a two-part fitting.

Non-tension tubular fittings used on PVC covered conductors shall allow the stripped end of the conductor and a quantity of insulation to be inserted into the tube. This shall be designed to reduce metal fatigue and prevent the ingress of moisture. The tube shall therefore exceed the minimum stripping length required and ensure that electrical performance is maintained.

Single holes in flat palms shall be of a medium fit series in accordance with BS EN 20273. Double hole palms shall be drilled in accordance with ENA TS 41-16, Fig. 4.

Bi-metal fittings shall be designed to eliminate any effect arising from galvanic corrosion that would impair the performance of the fitting.

Bail fittings designed to be used in conjunction with live line taps shall be manufactured from copper or aluminium rod, in conjunction with diameters to suit the associated live line tap, subject to a range of 8 mm to 12 mm. The bail shall be compressed, welded or brazed onto a tee joint. The dimension of the bail shall be that the horizontal portion is a minimum distance of 90 mm below the main conductor, and its length shall be a minimum of 175 mm. This will enable it to accept a splash (arcing) plate.

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Compression fittings shall be installed in accordance with CE Electric UK guidance note NSP/004/108 – Guidance on the Installation of Compression Joints.

3.3 Mechanical Fittings

Fittings shall be marked in accordance with ENA TS 43-92 Issue 4 Sections 4.3.

Split bolt connectors shall be capable of being held during installation to avoid kinking of the conductors to which they are applied. Installation torque values shall be marked on fittings in newton-metres (Nm). Split bolt connectors shall be manufactured from tin plated brass or similar thus removing the risk of electrolytic action between the fitting and copper or aluminium conductors.

Live line taps shall be manufactured from high strength, corrosion resistant silicon Bronze or similar. Steel pins, rivets or springs shall be manufactured from corrosion resistant stainless steel. Beaks on live line taps shall protrude from the body of the fitting to allow ease of location onto a bail fitting. The eye fitting and shank length shall be compatible with a positive grip live line head currently used within CE Electric UK. It must be capable of withstanding a minimum torque value of 32Nm.

The 400 Amp live line tap shall accommodate 14mm to 70mm Copper and the 800 Amp live line tap shall accommodate 70mm to 125mm Copper.

3.4 Testing

Testing requirements for compression and mechanical conductor fittings shall be in accordance with ENA TS 43-92 Issue 4 and BS EN 61284.

4. References

4.1 External Documentation

Reference	Title
ENA TS 43-92 Issue 4 2004	Overhead Line Fittings
BS EN 1461: 1999	Hot dipped galvanised coatings on fabricated iron or steel articles – specification and test methods
BS EN 20273: 1992	Fasteners – Clearance holes for bolts and screws
BS EN 61284: 1997	Overhead Lines – Requirements and tests for fittings

4.2 Internal documentation

Reference	Title
NSP/004/108: 2006	Guidance on the Installation of Compression Joints

5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	14/12/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Purchasing Operations Manager	G Bolden	20/12/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager	G Hammel	20/12/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	11/01/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	24/12/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	27/12/07

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APPENDIX A

SELF CERTIFICATION CONFORMANCE DECLARATION

Overhead line fittings covered by ENATS 43-92 shall comply with the latest issues of the relevant international and British Standards. ENATS 43-92 is intended to amplify and/or clarify the requirements of those Standards

This check sheet identifies the clauses in ENATS 43-92 and the clauses of the aforementioned Standards relevant to overhead line fittings. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENATS' as appropriate

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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Appendix A - Table 1

BS EN 61284			ENA TS 43-92			
Clause/Sub-clause	Requirements	Conformance Code	Clause/Sub-clause	Requirements	Conformance Code	
4	Requirements ¹		4	General requirements ²		
4.2	Requirements for specific fittings ³		5 6	Compression conductor fittings ⁴ Mechanical conductor fittings		
5	Quality Assurance					
6	Classification of Tests ⁵					

¹ Clause 4.2 of BS EN 61284 is subject to Clauses 5, 6 and 7 of ENATS 43-92. Refer to Table 1 Part 2.

² ENATS 43-92 calls for specific requirements in relation to;

- finish (ENATS 43-92 Clause 4.1.3)
- tolerances (ENATS 43-92 Clause 4.2)
- Packaging and protection (ENATS 43-92 Clause 4.4)
- Tension terminations (ENATS 43-92 Clause 4.5).

³ Clause 4.2 of BS EN 61284 covers general design requirements, the requirements of which shall be complied with, where appropriate to the fitting employed.

⁴ Clauses 5, 6 and 7 of ENATS 43-92 calls for specific design requirements of compression conductor fitting not called by Clause 4.2 of BS EN 61284.

⁵ Clause 8.3 of ENATS 43-92 defines sampling and acceptance criteria.

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Appendix A – Table 2

BS EN 61284			ENATS 43-92			Remarks
Clause/Sub-clause	Requirements	Conformance Code	Clause/Sub-clause	Requirements	Conformance Code	
7 8 9 10 11 13	Visual examination Dimensional and material variation – hot dip galvanising Non destructive testing Mechanical heat tests Heat cycle test		8	Quality Assurance ⁶		
12	Magnetic loss tests					

⁶ The appropriate clauses of BS EN 61284 apply with the following specific requirements:

- The heat cycle test for current carrying fittings shall be 1000 cycles to a maximum temperature rise of 70° above ambient.
- The heat cycle test for current carrying fittings shall be undertaken on six fittings subject to the requirements of Clause 8.1.1 of ENATS 43-92.
- Clauses 8.2.1, 8.2.2, 8.2.3, 8.2.4, of ENATS 43-92 define the conductor tensions to be applied during the mechanical tests and the holding period T.
- Clause 8.2.2 of ENATS 43-92 calls for additional requirements in terms of the reapplication of the fitting to the conductor.
- Clauses 8.2.5 and 8.2.6 of ENATS 43-92 calls for additional torque tests for split bolts, saddle fittings and live line taps.
- Clauses 8.2.7 of ENATS 43-88 calls for additional tests to prove the interface integrity of the friction welded fittings.
- Clauses 8.2.9 of ENATS 43-92 calls method B (vertical damage and failure load tests) for suspension clamps and defines specific application and slippage criteria.

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Appendix B – Schedule of Components

Commodity Code	Description
Test Slug	
307322	Compression Fitting, Test Slug for VP6 Overhead Compression Tool Ref. VPTS to drawing Number 1.09.145.0056 Sheet 4
Single Hole Lug Fitting	
075226	Compression Fitting, Lug for 13/35 mm Cu with M17 Hole Ref. VPCL 1.1 (17)
075234	Compression Fitting, Lug for 13/35 mm Cu with M21 Hole Ref. VPCL 1.1 (21)
268823	Compression Fitting, Lug for 13 to 35 mm Cu with M8 Hole Ref. VPCL 1.1/2 (8) to Drawing Number 1.09.101.0237 sheet 3
075218	Compression Fitting, Lug One Hole for 13/35 mm Cu with M12 Hole Ref. VPCL 1.1/2 (12)
268838	Compression Fitting, Lug for 14 to 35 mm Cu with M14 Hole Ref. VPCL 1.1/2 (14) to Drawing Number 1.09.101.0237 sheet 3
268842	Compression Fitting, Lug for 32 to 70 mm Cu with M14 Hole Ref. VPCL 1.3 (14) to Drawing Number 1.09.101.0237 sheet 3
268857	Compression Fitting, Lug for 32 to 70 mm Cu with M17 Hole Ref. VPCL 1.3 (17) to Drawing Number 1.09.101.0237 sheet 3
268876	Compression Fitting, Lug for 95 - 100 mm Cu with M14 Hole Ref. VPCL 1.4 (14) to Drawing Number 1.09.101.0237 sheet 3
268880	Compression Fitting, Lug for 95 - 100 mm Cu with M18 Hole Ref. VPCL 1.4 (18) to Drawing Number 1.09.101.0237 sheet 3
268895	Compression Fitting, Lug for 95 - 100 mm Cu with M23 Hole Ref. VPCL 1.4 (23) to Drawing Number 1.09.101.0237 sheet 3
268908	Compression Fitting, Lug for 125 mm Cu with M14 Hole Ref. VPCL 1.5 (14) to Drawing Number 1.09.101.0237 sheet 3
268912	Compression Fitting, Lug for 125 mm Cu with M18 Hole Ref. VPCL 1.5 (18) to Drawing Number 1.09.101.0237 sheet 3
268927	Compression Fitting, Lug for 125 mm Cu with M23 Hole Ref. VPCL 1.5 (23) to Drawing Number 1.09.101.0237 sheet 3
Double Hole Lug Fitting	
268950	Compression Fitting, Lug Double M14 Hole for 32 mm HDBC Ref. VPCL 2.1/2C to Drawing Number 1.00.043.9203
268772	Compression Fitting, Lug Double M14 Hole for 70 mm HDBC Ref. VPCL 2.3C to Drawing Number 1.00.043.9203
268965	Compression Fitting, Lug Double M14 Hole for 100 mm HDBC Ref. VPCL 2.4C to Drawing Number 1.00.043.9203
268984	Compression Fitting, Lug Double M14 Hole for 125 mm HDBC Ref. VPCL 2.5C to Drawing Number 1.00.043.9203
Aluminium Lug Fitting	
075192	Compression Fitting, Angle Lug for 13/35 mm Cu with M17 Hole Ref. VPCL 1.1 (17) 60
075127	Compression Fitting, Angle Lug for 13/35 mm Cu with M12 Hole Ref. VPCL 1.1/2 (14) 60
075143	Compression Fitting, Angle Lug for 32 to 70 mm Cu with M14 Hole Ref. VPCL 1.3 (14) 60
075374	Compression Fitting, Angle Lug for 32 to 70 mm Cu with M17 Hole Ref. VPCL 1.3 (17) 60
367588	Compression Fitting, Angle lug for 13/35 mm Cu with M17.5 Hole to Drawing Number 1.09.101.0237 Sheet 2
367573	Compression Fitting, Angle lug for 13/35 mm Cu with M14.3 Hole to Drawing Number 1.09.101.0237 Sheet 2

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Commodity Code	Description
Bi-metal Lug Fitting	
260107	Compression Fitting, Lug Bi-metal 50 - 95 mm ABC, 12.5 mm Hole Ref. VPBL 1.2 to Drawing Number 1 09 119 3157 sheet 3
240372	Compression Fitting, Lug Bi-metal 120 mm ABC, 12.5 mm Hole Ref. VPBL 1.3 to Drawing Number 1 09 119 3157 sheet 4
Copper Tension Fitting	
268397	Compression Fitting, Mid Span Tension for 13 - 32 mm Cu Ref. VPCT 1.2 to Drawing Number 1.09.101.0102 Sheet 3
072363	Compression Fitting, Mid Span Tension for 50 - 70 mm Cu Ref. VPCT 1.3 Drawing Number 1.09.101.0102 Sheet 3)
268429	Compression Fitting, Mid Span Tension for 70 - 100 mm Cu Ref. VPCT 1.4 to Drawing Number 1.09.101.0102 Sheet 3
268325	Compression Fitting, Mid Span Tension for 70 - 100 mm Cu Ref. VPCT 1.4 to Drawing Number 1.09.101.0102 Sheet 3
Aluminium Tension Fitting	
072405	Compression Fitting, Tension for 7.4mm - 11.3 mm Al Ref. VPAT 1.3
268363	Compression Fitting, Tension for 100 mm ACSR Ref. VPAT 1.3 to Drawing Number 1 09 101 0102 sheet 4
268378	Compression Fitting, Tension for 150 mm ACSR Ref. VPAT 1.4 to Drawing Number 1 09 101 0102 sheet 4
Copper Non-tension Fitting	
268452	Compression Fitting, Non-tension for 13 mm to 32 mm Cu Ref. VPCN 1.2 to Drawing Number 1.00.043.9203
072603	Compression Fitting, Non-tension for 32 mm to 70 mm Cu Ref. VPCN 1.3 to Drawing Number 1.00.043.9204
260624	Compression Fitting, Non-tension for 70 mm to 100 mm HDDB Ref. VPCN 1.4 to Drawing Number 1.00.043.9203
268471	Compression Fitting, Non-tension for 95 mm to 125 mm HDDB Ref. VPCN 1.5 to Drawing Number 1.00.043.9203
Pin Connector Fitting	
260622	Compression Fitting, Non-tension Set 6 x 70 mm Flexible Cu Jumper to Nulec Reclosers (c/w 2 x Heat Shrink Tubes/fitting) Ref. VPCN 1.8 to Drawing Number 1.09.101.0102 Sheet 13
260620	Compression Fitting, Set 6 x 32 mm HDDB to 70 MM Cu Flexible Ref. VPCN 1.9 to Drawing Number 1.00.043 9202 Sheet 4
Aluminium Non-tension Fitting	
072652	Compression Fitting, Non-Tension for 4mm to 25 mm Al Ref. VPAN 1.2
268433	Compression Fitting, Non-Tension for 50 mm ACSR and AA Ref. VPAN 1.2 to Drawing Number 1 00 043 9204 sheet 2
268448	Compression Fitting, Non-Tension for 100 mm ACSR and AA Ref. VPAN 1.3 to Drawing Number 1 00 043 9204 sheet 3
Bi-metal Non-tension Fitting	
246932	Compression Fitting, Non-tension Bi-metal 50/100 mm AA to 13/32 mm Cu Ref. VPBN 1.1 to Drawing Number 1 09 101 0649 sheet 2
268518	Compression Fitting, Non-tension Bi-metal 50/100 mm Al to 35/70 mm HDDB Ref. VPBN 1.2 to Drawing Number 1 09 101 0649 sheet 2
268503	Compression Fitting, Non-tension Bi-metal 100/175 mm Al to 70/100 mm HDDB Ref. VPBN 1.3 to Drawing Number 1 09 101 0649 sheet 2

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Commodity Code	Description
071365	Compression Fitting, Non-tension Bi-metal 100/175 mm Al to 100/120 mm HDBC Ref. VPBN 1.5
071381	Compression Fitting, Non-tension Bi-metal, 120 deg. 35/70 mm Cu to 70 mm Al Ref. VPBN 3.10
Tap Connector Fitting	
077370	Compression Fitting, Tap Connector Main 10 - 35 mm, Tap 10 - 35 mm Ref. VPCTC 2.3
077347	Compression Fitting, T Connector for 10 - 38 mm Cu Main to 32 - 150 mm Cu Tap Ref. VPCTC 3.2
077354	Compression Fitting, T Connector for 25 - 70 mm Cu Main to 25 - 70 mm Cu Tap Ref. VPCTC 3.3
268804	Compression Fitting, T Connector for 32 - 150 mm Cu Main to 50/95 mm ABC or 95 mm HDBC Tap Ref. VPCTC 3.2 to Drawing Number 1.09.119.3151 Sheet 3
077362	Compression Fitting, Tap Connector for 25 - 70 mm Cu Main, 10 - 35 mm Cu Tap Ref. VPCSC 2.2
268787	Compression Fitting, Tee Connector for 10 - 60 mm Al Main to 10 - 60 mm Al Tap. Ref. VPATC 3.1 to Drawing Number 1 09 119 3151 sheet 2
268791	Compression Fitting, Tee Connector for 22 - 100 mm Al Main to 22 - 100 mm Al Tap. Ref. VPATC 3.3 to Drawing Number 1 09 119 3151 sheet 2
077453	Compression Fitting, Service Connector for 22 - 100 mm Main to 10 - 60 mm Tap Al. Ref. VPASC 2.2
077438	Compression Fitting, Bi-metal Tap for 32 - 95 mm Cu Main to 22 - 100 mm Al Tap Ref. VPBT 1.2
077461	Compression Fitting, Bi-metal for 22 - 100 mm Al Main to 10 - 35 mm Cu Tap Ref. VPBT 2.2
Bail Fittings	
268630	Compression Fitting, Stirrup Bail Clamp for 13 to 32 mm Cu Ref. VPCS 1.1 to Drawing Number 1.09.120.0013
268645	Compression Fitting, Stirrup Bail Clamp for 70 to 125 mm HDBC Ref. VPCS 1.2 to Drawing Number 1.09.120.0013
268537	Compression Fitting, Stirrup Bail Clamp for 50 mm ACSR and AAAC Ref. VPAS 1.2
268541	Compression Fitting, Stirrup Bail Clamp for 100 mm ACSR and AAAC Ref. VPAS 1.3
241892	Compression Fitting, Stirrup Bail Clamp for 150 - 175 mm ACSR and AAAC Ref. VPAS 1.4
246951	Compression Fitting, Spill Take Off, Bi-metal for Heat Shrink Cable Connections Ref. VPST 1.1 to Drawing Number 1.09.101.0649
ACSR Fitting	
245709	Compression Fitting, Anchor Clamp for 150/175 mm ACSR to Drawing Number 1 09 101 0653 sheet 1
245747	Compression Fitting, Anchor Clamp for 60/70 mm ACSR to Drawing Number 1.09.101.0653 Sheet 1
245728	Compression Fitting, Tension for 150/175 mm ACSR to Drawing Number 1 09 101 0653 sheet 1
245925	Compression Fitting, Tension for 60/70 mm ACSR to Drawing Number 1 09 101 0653 sheet 1
245713	Compression Fitting, Angle Lug for 150/175 mm ACSR to Drawing Number 1 09 101 0653 sheet 1
245836	Compression Fitting, Jumper Lug for 60/70 mm ACSR to Drawing Number 1.09.101.0653 Sheet 1
245732	Compression Fitting, Lug Bi-metal for 125 mm Cu to Drawing Number 1.09.101.0653 Sheet 1

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Commodity Code	Description
AAAC Fitting	
243240	Compression Fitting, Anchor Clamp for 200mm AAAC with Bolt Plate to Drawing Number 1 09 101 0653 sheet 3
243255	Compression Fitting, Anchor Clamp for 175 mm AAAC, with Eye Termination End Fitting to Drawing Number 1 09 101 0653 sheet 2
243239	Compression Fitting, Anchor Clamp for 175 mm AAAC, Slotted Palm and Bolt Plate to Drawing Number 1 09 101 0653 sheet 2
243224	Compression Fitting, Tension for 175 mm AAAC to Drawing Number 1 09 101 0653 sheet 2
243243	Compression Fitting, Angle Lug for 175 mm AAAC to Drawing Number 1 09 101 0653 sheet 2
240781	Compression Fitting, Lug for 175 mm AAAC to Drawing Number 1 09 101 0653 sheet 2
243281	Compression Fitting, Lug Angle Bi-metal for 125 mm HDBC 175 mm AAAC to Drawing Number 1 09 101 0653 sheet 2
240885	Compression Fitting, Lug for 125 mm HDBC - 175 AAAC to Drawing Number 1.09.101.0653 Sheet 2
Anchor Clamp and Lug Fitting	
247827	Compression Fitting, Anchor Clamp for 50 mm AAAC/ACSR Ref. VPAA 2.2 (S) SL, to Drawing Number 1 09 101 0102 sheet 10
268274	Compression Fitting, Lug Angle with Bolt Plate for 50 mm ACSR/AAAC Ref. VPAL 2.2 (A)N to Drawing Number 1 09 1010102 sheet 9
247899	Compression Fitting, Anchor Clamp for 100 mm AAAC/ACSR Ref. VPAA 2.3(S) SL, to Drawing Number 1 09 101 0102 sheet 11
268293	Compression Fitting, Lug Angle with Bolt Plate for 100 mm AAAC and 100/150/175 ACSR Ref. VPAL 2.4 (A) N to Drawing Number 1 09 101 0102 sheet 5
248162	Compression Fitting, Anchor Clamp for 150 mm ACSR Ref. VPAA 2.4 (S) SL, to Drawing Number 1 09 101 0102 sheet 9
247935	Compression Fitting, Anchor Clamp for 13 mm Cad and 32 mm HDBC Ref. VPCA 2.2 (S)SL to Drawing Number 1.09.101.0102 Sheet 9
248444	Compression Fitting, Lug Angle with Bolt Plate for 13mm Cad and 32 mm HDBC Ref. VPAL 2.1/2 (A) N to Drawing Number 1 09 101 0102 sheet 5
247920	Compression Fitting, Anchor Clamp for 70 to 100 mm HDBC Ref. VPCA 2.4(S)SL to Drawing Number 1.09.101.0102 Sheet 9
248529	Compression Fitting, Lug Double M14 Hole for 100 mm HDBC Ref. VPCL 2.4(A)N to Drawing Number 1.09.101.102 Sheet 9
247916	Compression Fitting, Anchor Clamp for 125 mm HDBC Ref. VPCA 2.5 (S)SL to Drawing Number 1.09.101.0102 Sheet 9
248393	Compression Fitting, Lug Double M14 Hole for 125 mm HDBC Ref. VPCL 2.5(A)N to Drawing Number 1.09.101.102 Sheet 9
247831	Compression Fitting, Lug Angle for 100 mm AAAC and 100/150/175 ACSR Ref. to Drawing Number 1 09 101 0102 sheet 5
261951	Compression Fitting, Lug Double Hole for 32 mm HDBC/50 mm AA Ref. VPAL 2.2 (A) to Drawing Number 1 09 101 0102 sheet 5
261966	Compression Fitting, Lug for 70-100 mm Cu PVC and 100 mm AA Ref. VPAL 2.4(C) to Drawing Number 1 09 101 0102 sheet 9
235764	Compression Fitting, Anchor Clamp for 50 - 120 mm XLPE Ref. VPAA 7.3 SDCC to Drawing Number 1.09.101.0667 Sheet 1
245982	Compression Fitting, Socket Anchor Clamp for 175 mm AAAC to Drawing Number 1.09.101.0653 Sheet 3

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Commodity Code	Description
246059	Compression Fitting, Ball Anchor Clamp for 175 mm AAAC to Drawing Number 1.09.101.0653 Sheet 3
Jumper Lugs for ACSR and HDBC Fittings – Alcan Pattern	
246415	Compression Fitting, Alcan Male Lug for 150 mm ACSR to Drawing Number 1.09.101.0420 Sheet 4
246434	Compression Fitting, Alcan Male Lug for 175 mm ACSR to Drawing Number 1.09.101.0420 Sheet 4
246379	Compression Fitting, Alcan Female Lug for 60 mm ACSR to Drawing Number 1.09.101.0420 Sheet 4
246383	Compression Fitting, Alcan Female Lug for 70 mm ACSR to Drawing Number 1.09.101.0420 Sheet 4
246398	Compression Fitting, Alcan Female Lug for 150 mm ACSR to Drawing Number 1.09.101.0420 Sheet 4
246400	Compression Fitting, Alcan Female Lug, for 175 mm ACSR to Drawing Number 1.09.101.0420 Sheet 4
266620	Compression Fitting, Alcan Female Lug, Bi-metal for 125 mm HDBC to Drawing Number 1.09.101.0420 Sheet 4
241182	Compression Fitting, Alcan Female Lug, Bi-metal for 125mm PVC HDBC to Drawing Number 1.09.101.0420 Sheet 4
248783	Compression Fitting, Alcan Female Angle Lug, for 150 mm ACSR to Drawing Number 1.09.101.0420 Sheet 4
248798	Compression Fitting, Alcan Female Angle Lug, Non Bi-metal for 175 mm ACSR to Drawing Number 1.09.101.0420 Sheet 4
266635	Compression Fitting, Alcan Female Angle Lug, Bi-metal for 125 mm HDBC to Drawing Number 1.09.101.0420 Sheet 4
241178	Compression Fitting, Angle Lug, Bi-metal for 125 mm PVC HDBC to Drawing Number 1.09.101.0420 Sheet 4
246472	Compression Fitting, Jumper Dead End for 60/70 mm ACSR to Drawing Number 1 09 101 0420 sheet 9
246925	Compression Fitting, Tension Joint for 60/70 mm ACSR to Drawing Number 1.09.101.0653 Sheet 2
247545	Compression Fitting, Tee with Jumper Lug for 13 - 32 mm Cu Ref. VPCTC 4.1 to Drawing Number 1.09.101.0102 Sheet 10
247597	Compression Fitting, Tee with Jumper Lug for 70, 100 and 125 mm HDBC Ref. VPCTC 4.4 to Drawing Number 1.09.101.0102 Sheet 10
247530	Compression Fitting, Tee for 175 mm AAAC and 150 mm ACSR Ref. VPATC 4.5 to Drawing Number 1.09.101.0102 Sheet 10
247333	Compression Fitting, Tee for 50 - 100 mm AAAC and ACSR Ref. VPATC 4.3 to Drawing Number 1.09.101.0102 Sheet 10
246364	Compression Fitting, Tee Connector for 175 mm ACSR to Drawing Number 1.09.101.0420 Sheet 4 or 0102 Sheet 10
241159	Compression Fitting, Tension for Double Jumper 150 mm ACSR to Drawing Number 1.09.101.0420 Sheet 11
241125	Compression Fitting, Tension for Double Jumper 175 mm ACSR to Drawing Number 1.09.101.0420 Sheet 11
Split Bolt Connector Fittings	
266480	Connector, Split Bolt Tin Plated for 16mm Copper to Drawing Number 1.09.119.3100 Sheet 2
266599	Connector, Split Bolt Tin Plated for 32mm Copper to Drawing Number 1.09.119.3100 Sheet 2

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266601	Connector, Split Bolt Tin Plated for 70mm Copper to Drawing Number 1.09.119.3100 Sheet 2
266616	Connector, Split Bolt Tin Plated for 100mm Copper to Drawing Number 1.09.119.3100 Sheet 2
Commodity Code	Description
241438	Connector, Split Bolt Tin Plated for 120mm Copper to Drawing Number 1.09.119.3100 Sheet 2
Live Line Tap Fittings	
268999	Live Line Tap, 400 Amp utilised at 200 Amp to Drawing Number 1.09.121.0024
269008	Live Line Tap, 800 Amp utilised at 400 Amp to Drawing Number 1.09.121.0024